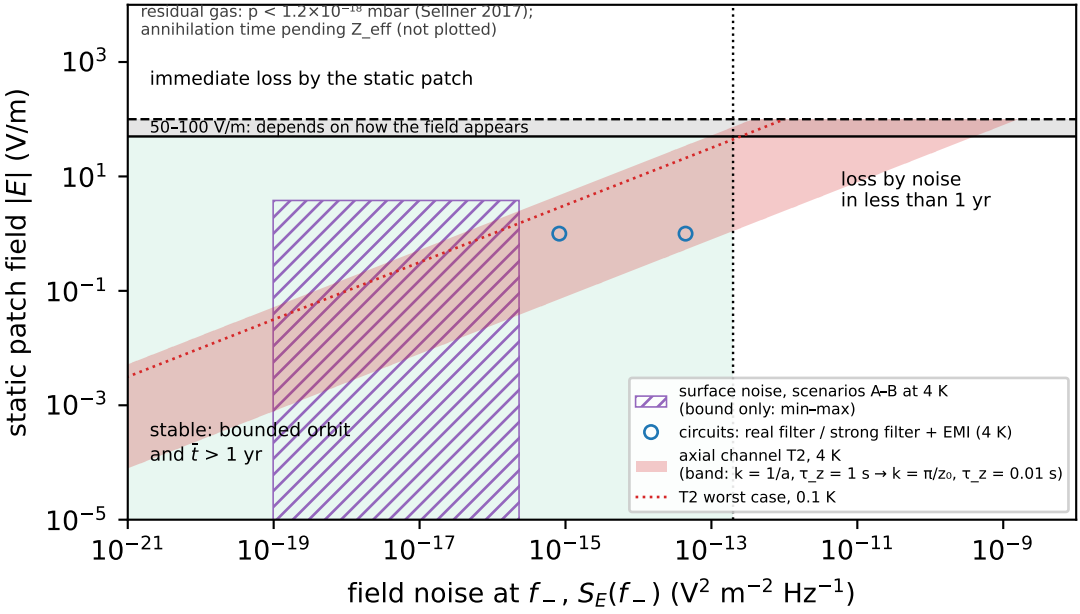
residual gas: p < 1.2×10⁻¹⁸ mbar (Sellner 2017);
annihilation time pending Z_eff (not plotted)
immediate loss by the static patch
50–100 V/m: depends on how the field appears
loss by noise
in less than 1 yr
stable: bounded orbit
and t̄ > 1 yr
surface noise, scenarios A–B at 4 K
(bound only: min–max)
circuits: real filter / strong filter + EMI (4 K)
axial channel T2, 4 K
(band: k = 1/a, τ_z = 1 s → k = π/z₀, τ_z = 0.01 s)
T2 worst case, 0.1 K
static patch field |E| (V/m)
field noise at f_− , S_E(f_−) (V² m⁻² Hz⁻¹)
10³
10¹
10⁻¹
10⁻³
10⁻⁵
10⁻²¹
10⁻¹⁹
10⁻¹⁷
10⁻¹⁵
10⁻¹³
10⁻¹¹
10⁻⁹